

J. Zachary Gazak

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Research Interests

Vision-language model interpretability, spatial reasoning in multimodal models, computer vision, autonomous sensing systems, object detection and density estimation, Bayesian uncertainty quantification, space domain awareness, astronomical instrumentation, stellar spectroscopy.

Education

Ph.D., Astronomy · University of Hawai'i at Mānoa · 2014 Advisor: Rolf-Peter Kudritzki. Thesis on quantitative near-infrared spectroscopy of red supergiant stars as extragalactic metallicity probes.

M.S., Astronomy · University of Hawai'i at Mānoa · 2010

B.S., Physics · Massachusetts Institute of Technology · 2006

Professional Experience

Principal Investigator & Astronomer · Space Systems Command (SSC), Kihei, HI · 2020–Present PI of the Machine Intelligence for Space Superiority lab at U.S. Space Systems Command. Lead a 10-person, multi-organization research team (KBR, Boeing, EOS, MIT, Virginia Tech, Macquarie University, UNSW) developing autonomous ML systems for operational deployment in high-stakes space domain awareness environments. Designed and deployed novel sensing modalities (spectroscopic, neuromorphic, multispectral) with end-to-end pipelines from data collection through model training to operational deployment across distributed autonomous sensor networks.

Independent Researcher · Sidereal LLC, Haiku, HI · 2024–Present Independent ML research targeting top-tier venues (NeurIPS, ICLR). Current program: diagnosing and mechanistically characterizing spatial reasoning failures in vision-language models via the Sophon benchmark and Frozen Capability Audit framework.

Research Physicist · Air Force Research Laboratory (AFRL), Kihei, HI · 2019–2020 Founded a new ML research program for space domain awareness. Built end-to-end simulation, training, and validation pipelines for learned classification from sensor data. Revitalized legacy sensor hardware into high-yield operational platforms through software modernization.

Technology Co-Founder · Alfie Hospitality, Sydney, Australia · 2018–2019 Designed, built, and deployed full-stack backend services, mobile applications (iOS/Android), and venue-facing web tools for commercial hospitality operations.

Venture Capital Associate · H2 Ventures, Sydney, Australia · 2016–2018 Directed accelerator programs; mentored early-stage fintechs including Weel (DiviPay, \$20M Series A), Forage (\$25M Series B), and Spriggy (\$35M Series B, NAB Ventures).

Data Scientist & Program Director · Insight Data Science, Palo Alto, CA · 2014–2016 Led fellowship programs training PhDs for industry data science roles. Founded Insight's internal data science team supporting rapid organizational growth.

Graduate Research Assistant & Postdoctoral Researcher · University of Hawai‘i at Mānoa · 2008–2014 Developed automated spectroscopic analysis techniques for extragalactic red supergiant stars. Published first-author papers in leading astrophysics journals. Awarded competitive observing time on Keck, Subaru, and IRTF telescopes.

Publications

2,200+ citations across 55+ publications (15 first-authored). Full list: [Google Scholar](#).

Accepted / Submitted

1. **J. Z. Gazak**, R. Swindle, J. Fletcher, “Sophon: A Procedural Diagnostic for Spatial Reasoning in Vision-Language Models,” NeurIPS 2026, Datasets & Benchmarks Track (accepted).
2. **J. Z. Gazak**, R. Swindle, L. Spitler, J. Fletcher, “AllClear: Blind All-Sky Astrometric Calibration and Directional Sky-Transparency Maps,” submitted to Publications of the Astronomical Society of the Pacific (2026).

Peer-Reviewed Journal & Conference Papers (First-Authored)

3. **J. Z. Gazak**, R. Swindle, S. Morales, K. Iott, E. Blackhurst, J. Fletcher, “Rapid Deployment, Calibration, and Training of Optical Observatories for Space Domain Awareness,” Journal of the Astronautical Sciences (JASS), 2026 (accepted).
4. **J. Z. Gazak**, L. Fisher, M. Phelps, R. Swindle, L. Baruela, J. Fletcher, “SENPAl: Sidereal Enriched Rate-Track Astrometry in Deep Imagery of Solar System Bodies,” Publications of the Astronomical Society of the Pacific, 2026. DOI: [10.1088/1538-3873/ae2b35](https://doi.org/10.1088/1538-3873/ae2b35).
5. **J. Z. Gazak**, I. McQuaid, R. Swindle, M. Phelps, J. Fletcher, “SpectraNet: Learned Recognition of Artificial Satellites from High Contrast Spectroscopic Imagery,” IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2022, pp. 2403–2411.
6. **J. Z. Gazak**, R.-P. Kudritzki, C. Evans, L. Patrick, B. Davies, M. Bergemann, B. Plez, F. Bresolin, R. Bender, M. Wegner, A. Z. Bonanos, S. J. Williams, “Red Supergiants as Cosmic Abundance Probes: The Sculptor Galaxy NGC 300,” The Astrophysical Journal, 805, 182, 2015.
7. **J. Z. Gazak**, B. Davies, N. Bastian, R.-P. Kudritzki, M. Bergemann, B. Plez, C. Evans, L. Patrick, F. Bresolin, E. Schinnerer, “A New Method for Measuring Metallicities of Young Super Star Clusters,” The Astrophysical Journal, 787, 142, 2014.
8. **J. Z. Gazak**, B. Davies, R.-P. Kudritzki, M. Bergemann, B. Plez, “Quantitative Spectroscopic J-band Study of Red Supergiants in Perseus OB-1,” The Astrophysical Journal, 788, 58, 2014.
9. **J. Z. Gazak**, N. Bastian, R.-P. Kudritzki, A. Adamo, B. Davies, B. Plez, M. A. Urbaneja, “The Red Supergiants in the Super Star Clusters of M83,” Monthly Notices of the Royal Astronomical Society Letters, 430, L35, 2013.
10. **J. Z. Gazak**, J. A. Johnson, J. Tonry, D. Dragomir, J. Eastman, A. W. Mann, E. Agol, “Transit Analysis Package: An IDL Graphical User Interface for Exoplanet Transit Photometry,” Advances in Astronomy, 2012. (100+ citations)

Proceedings & Technical Reports (First-Authored)

11. **J. Z. Gazak**, R. Swindle, S. Morales, M. Phelps, K. Iott, E. Blackhurst, J. Fletcher, “Rapid Deployment, Calibration, and Training of Optical Observatories for Space Domain Awareness,” AMOS Conference, 2025. DOI: [10.64861/XGPY2184](https://doi.org/10.64861/XGPY2184).

12. **J. Z. Gazak**, R. Swindle, M. Phelps, J. Fletcher, "Simultaneous Detection, Recognition, and Localization of Geosynchronous Satellites from Ground Based Imagery," AMOS Conference, 2023.
13. **J. Z. Gazak**, R. Swindle, M. Phelps, I. McQuaid, J. Fletcher, "Learned Satellite Identification Using Low-Resolution Spectropolarimetry," Proc. SPIE 12112, 2022.
14. **J. Z. Gazak**, I. McQuaid, B. Wolfson, J. Fletcher, "Incremental Learning of Novel Resident Space Object Spectral Fingerprints," AMOS Conference, 2021.
15. **J. Z. Gazak**, J. Fletcher, R. Swindle, I. McQuaid, "Exploiting Spatial Information in Raw Spectroscopic Imagery using Convolutional Neural Networks," AMOS Conference, 2020.

Selected Co-Authored Publications

- M. Bergemann, R.-P. Kudritzki, **J. Z. Gazak**, B. Davies, B. Plez, "Red Supergiant Stars as Cosmic Abundance Probes: NLTE Effects in J-band Magnesium Lines," *The Astrophysical Journal*, 804, 113, 2015.
- C. Lardo, B. Davies, R.-P. Kudritzki, **J. Z. Gazak**, C. J. Evans, L. R. Patrick, M. Bergemann, B. Plez, "Red Supergiants as Cosmic Abundance Probes: The First Direct Metallicity Determination of NGC 4038 in the Antennae," *The Astrophysical Journal*, 812, 160, 2015.
- L. R. Patrick, C. J. Evans, B. Davies, R.-P. Kudritzki, **J. Z. Gazak**, M. Bergemann, B. Plez, A. M. N. Ferguson, "Red Supergiant Stars as Cosmic Abundance Probes: KMOS Observations in NGC 6822," *The Astrophysical Journal*, 803, 14, 2015.
- B. Davies, R.-P. Kudritzki, **J. Z. Gazak**, B. Plez, M. Bergemann, C. Evans, L. Patrick, "Red Supergiants as Cosmic Abundance Probes: Metallicities of M31," *The Astrophysical Journal*, 806, 21, 2015.
- P. S. Muirhead et al. (incl. **J. Z. Gazak**), "Kepler-445, Kepler-446 and the Occurrence of Compact Multiples Orbiting Mid-M Dwarf Stars," *The Astrophysical Journal*, 801, 18, 2015.
- C. Beichman et al. (incl. **J. Z. Gazak**), "Observations of Transiting Exoplanets with the James Webb Space Telescope (JWST)," *Publications of the Astronomical Society of the Pacific*, 126, 1134, 2014.
- M. Bergemann, R.-P. Kudritzki, M. Würl, B. Plez, B. Davies, **J. Z. Gazak**, "Red Supergiant Stars as Cosmic Abundance Probes: NLTE Effects in J-band Silicon Lines," *The Astrophysical Journal*, 764, 115, 2013.
- M. Bergemann, R.-P. Kudritzki, B. Plez, B. Davies, K. Lind, **J. Z. Gazak**, "Red Supergiant Stars as Cosmic Abundance Probes: NLTE Effects in J-band Iron and Titanium Lines," *The Astrophysical Journal*, 751, 156, 2012.
- B. J. Fulton et al. (incl. **J. Z. Gazak**), "Long-term Transit Timing Monitoring and Refined Light Curve Parameters of HAT-P-13b," *The Astronomical Journal*, 142, 84, 2011.
- J. N. Winn et al. (incl. **J. Z. Gazak**), "The Transit Ingress and the Tilted Orbit of the Extraordinarily Eccentric Exoplanet HD 80606b," *The Astrophysical Journal*, 703, 2091, 2009.

Selected Co-Authored Proceedings & Technical Reports

- E. De Alba, M. de Lannoy Kobayashi, A. Cabello, **J. Z. Gazak**, J. Fletcher, "Toward Integration of Large Language Models for Space Domain Awareness Command and Control," AMOS Conference, 2025.
- K. Chang, L. Fisher, **J. Z. Gazak**, J. Fletcher, "Resolved Hyperspectral Imaging," AMOS Conference, 2024.

A. Cabello, J. Houchard, C. Harris, **J. Z. Gazak**, J. Kadan, J. Fletcher, “Learned Initial Orbit Determination from Simulated Electro-Optical Observations,” AMOS Conference, 2024.

M. Phelps, T. Swindle, **J. Z. Gazak**, J. Fletcher, A. Vandenberg, “Improving Spectral-Based Estimation of Space Object Orientation,” AMOS Conference, 2022.

M. Phelps, **J. Z. Gazak**, T. Swindle, J. Fletcher, I. McQuaid, “Inferring Space Object Orientation with Spectroscopy and Convolutional Networks,” AMOS Conference, 2021.

Additional 30+ co-authored publications in ApJ, AJ, MNRAS, PASP, SPIE, and conference proceedings; see [Google Scholar](#) for the complete list.

Open-Source Software & Tools

Sophon — Procedurally generated benchmark for diagnosing spatial reasoning in vision-language models. Python package on PyPI (`sophon-bench`). [[github.com/zgazak](#) — private during review]

AllClear — Blind all-sky astrometric solver and directional sky-transparency mapper for autonomous observatory operations. [github.com/zgazak/allclear](#)

SENPAI — Sidereal Enriched Rate-Track Astrometry pipeline; sensor-agnostic, zero-calibration design for autonomous star/satellite detection. Python package on PyPI (`astro-senpai`). [github.com/ssc-ai/senpai](#)

SDASim — Speed-optimized differentiable satellite scene simulator for synthetic training data generation. PyTorch-based. [github.com/zgazak/sdasim](#)

Burr — Autonomous telescope calibration and observation scheduling system; integrates SENPAI and StarCSP for rapid-deploy SDA telescopes. [github.com/zgazak/burr](#)

StarCSP — Crowd-counting-inspired CNN for rapid star streak detection; trained with weak supervision from SENPAI ground truth on as few as one labeled streak per image. Released as GOTS.

astroeasy — Containerized astrometry.net plate solving with Python API. [github.com/ssc-ai/astroeasy](#)

TAP — Transit Analysis Package for exoplanet light curve analysis using Bayesian MCMC. Widely adopted in the exoplanet community; 100+ citations. [github.com/zgazak/TAP](#)

ORIENT — Orientation-Injected Encoding for Robust Neural Transformers; Gabor wavelet injection into CLIP ViT for encoder-level spatial improvements. [github.com/zgazak/orient](#)

Astra — Investigating what star streaks teach vision models about orientation and shape. [github.com/zgazak/astra](#)

Peer Review & Professional Service

Reviewer: CVPR (2023–present), ECCV (2024–present), WACV (2021–2023), AMOS (2020–present)

Session Chair & Invited Speaker: 5+ invited talks at AMOS Conference; 4+ SPIE presentations; WACV oral presentation; numerous internal DoD conferences and stakeholder briefings.

Technical Skills

Machine Learning & Interpretability: Mechanistic interpretability (linear probes, attention attribution, Jacobian analysis, residual-stream interventions, activation patching), LoRA fine-tuning and placement grids, frozen capability audits, computer vision (CNNs, object detection, density estimation / crowd counting), Bayesian inference (MC dropout, SWA/SWAG ensembles), weak supervision, distribution shift robustness, synthetic data generation. PyTorch, scikit-learn.

Vision-Language Models: Architecture-level diagnostics across LLaVA-NeXT, LLaVA-OneVision, Qwen2.5-VL, Idefics3, InternVL2.5; frontier model evaluation (GPT-4o, Gemini, Claude); cross-architecture comparative analysis; encoder and decoder probe pipelines.

Software & Infrastructure: Scientific Python, distributed training (multi-GPU, 2× A6000 48GB), experiment management, LLM API integration and prompt engineering, full-stack development (backend services, mobile, web), cloud infrastructure (AWS), serverless architecture, containerization (Docker), Git, CI/CD.

Astronomical Instrumentation: Optical/NIR spectroscopy, spectropolarimetry (LoVIS on AEOS 3.6m), telescope automation, astrometric calibration, plate solving, all-sky imaging, neuromorphic and multispectral sensor deployment. Observing experience on Keck (10m), Subaru (8.2m), IRTF (3m), VLT/KMOS (8.2m), AEOS (3.6m), and multiple 0.35–0.5m commercial telescopes.

Leadership & Communication: Technical strategy, cross-organization program management without direct authority, multi-stakeholder coordination (DoD, academia, industry), scientific writing, peer review for top-tier ML and astronomy venues.